

Zack and Luis

MIRC-F

Multimodal Information Retrieval
Challenge - Failed

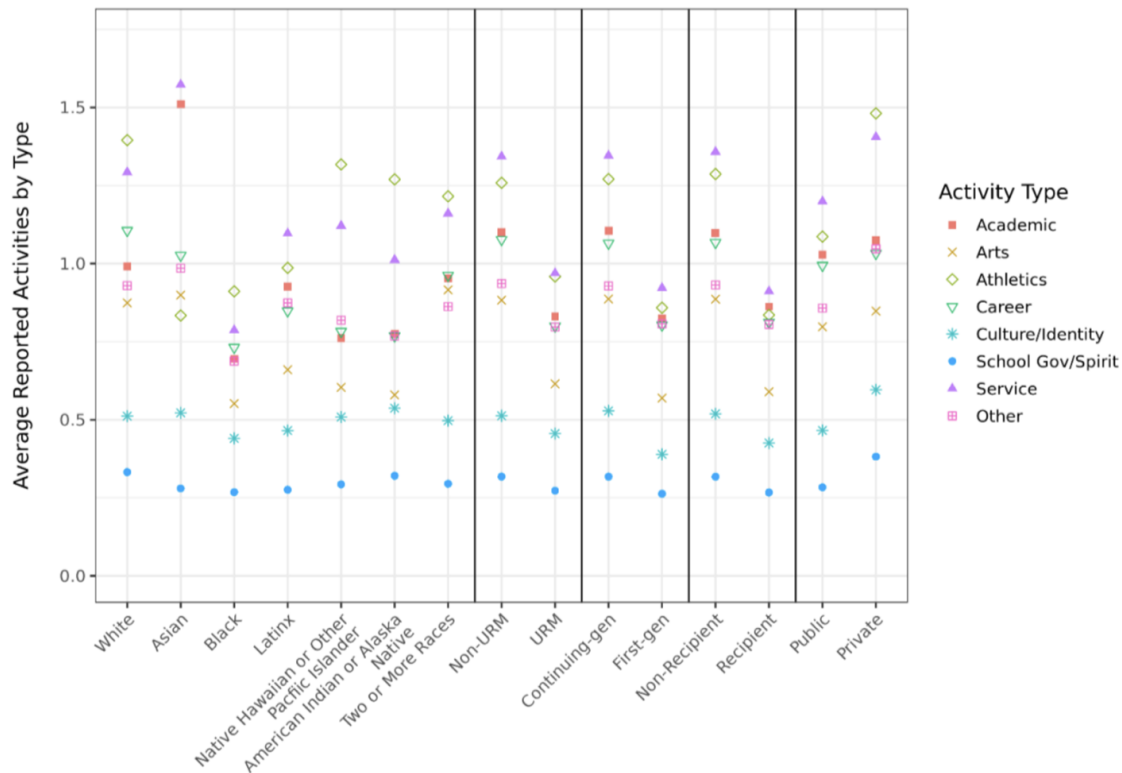


Image Credit: Fifth Dimension Team

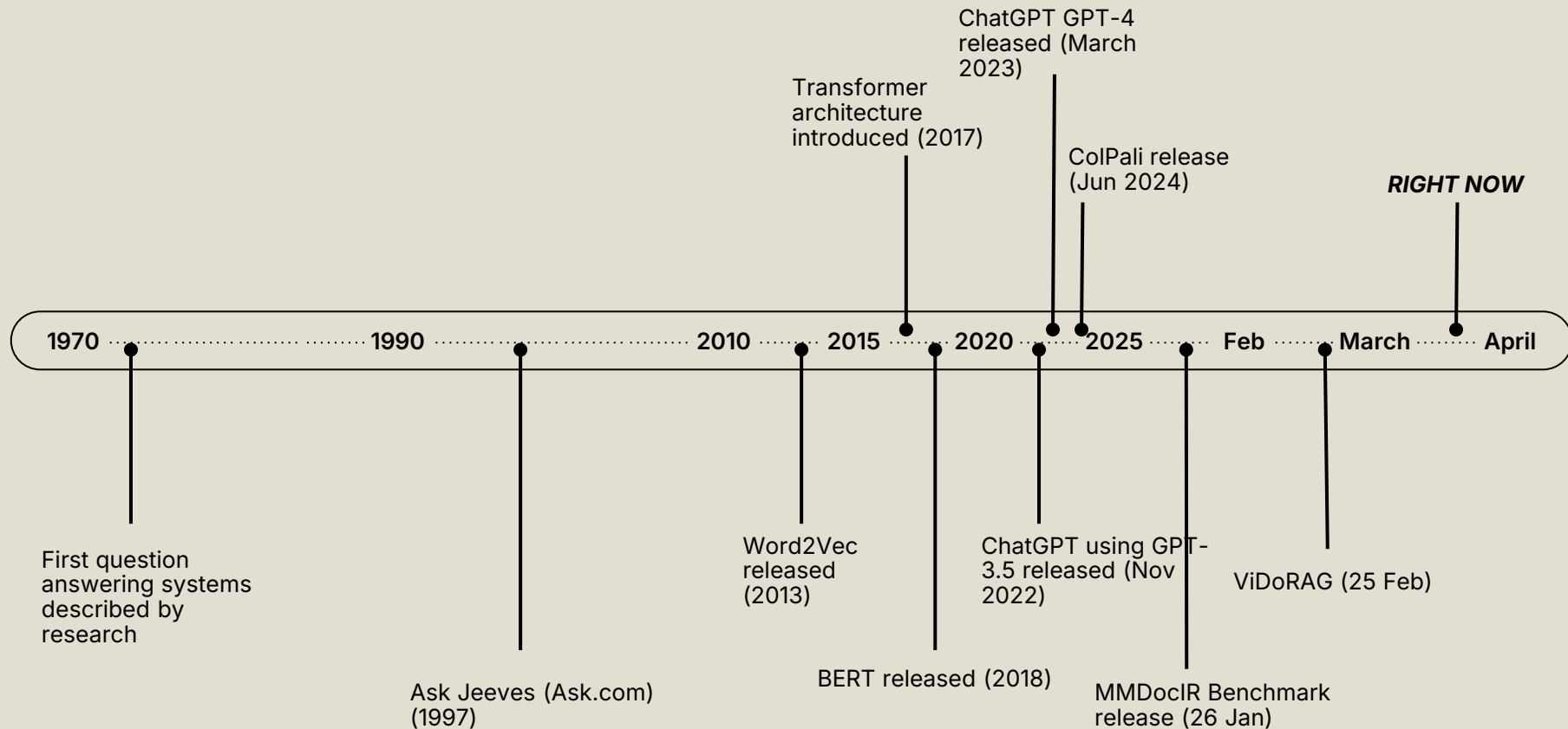
Text only?

Which demographic group reportedly participates in the most athletics?

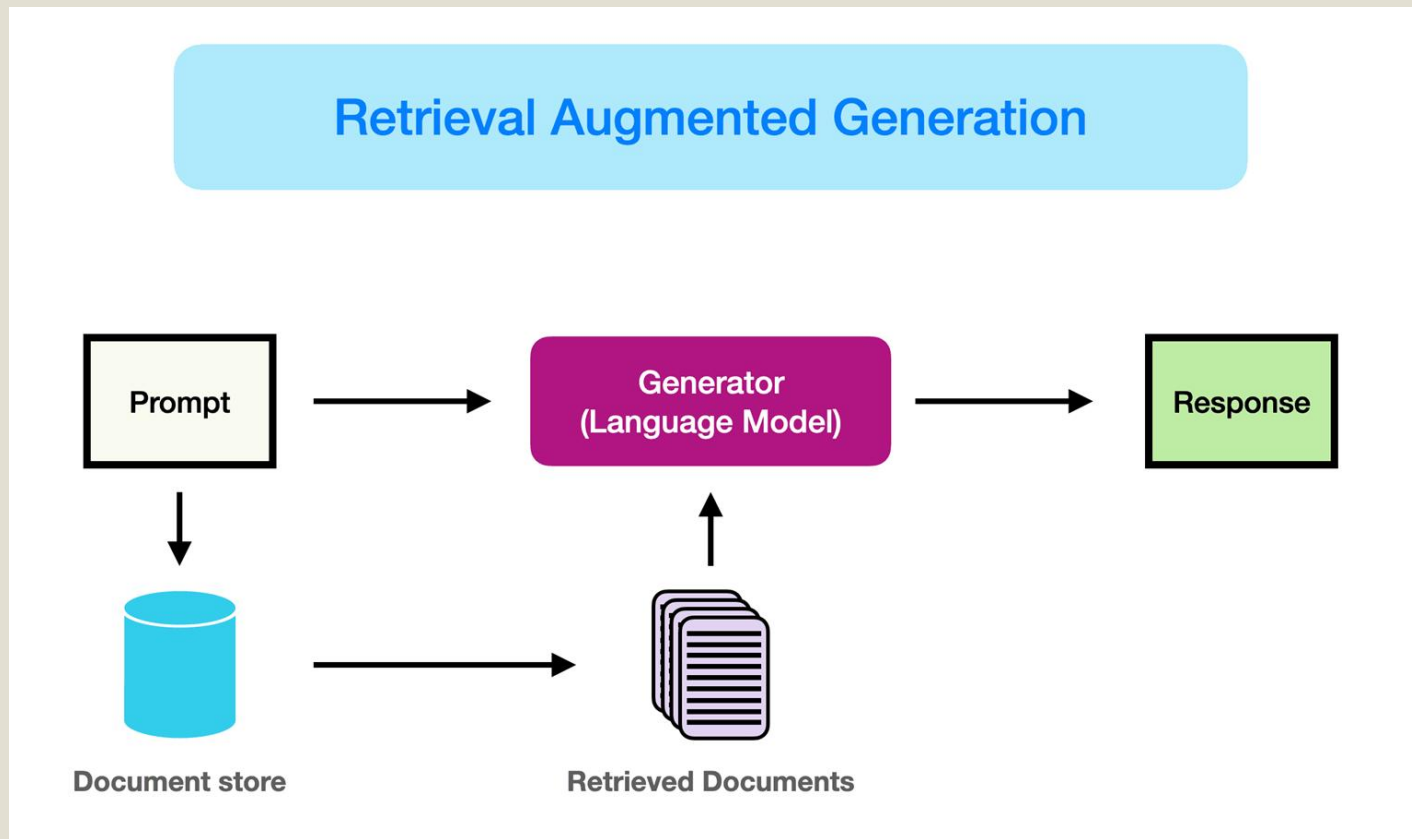
Figure 2. Average Number of Activities for Activity Type by Applicant Demographics



**WHAT IS MIR,
RAG, NLP?**



What is RAG?



Net Debt Ratio Total debt less cash and cash equivalents and marketable securities as a percentage of total debt less cash and cash equivalents and marketable securities, plus Chevron Corporation Stockholders' Equity, which indicates the company's leverage, net of its cash balances.

Millions of dollars	At December 31		
	2021	2020	2019
Short-term debt	\$ 256	\$ 1,548	\$ 3,282
Long-term debt	31,113	42,767	23,691
Total Debt	31,369	44,315	26,973
Less: Cash and cash equivalents	5,640	5,596	5,686
Less: Marketable securities	35	31	63
Total adjusted debt	25,694	38,688	21,224
Total Chevron Corporation Stockholders' Equity	139,067	131,688	144,213
Total adjusted debt plus total Chevron Corporation Stockholders' Equity	\$ 164,761	\$ 170,376	\$ 165,437
Net Debt Ratio	15.6 %	22.7 %	12.8 %

Capital Employed The sum of Chevron Corporation Stockholders' Equity, total debt and noncontrolling interests, which represents the net investment in the business.

Millions of dollars
Chevron Corporation Stockholders' Equity
Plus: Short-term debt
Plus: Long-term debt
Plus: Noncontrolling interest
Capital Employed at December 31

Return on Average Capital Employed (ROCE) and noncontrolling interest) divided by average sum of capital employed at the beginning and percentage of historical investments in the busi

Millions of dollars
Net income attributable to Chevron
Plus: After-tax interest and debt expense
Plus: Noncontrolling interest
Net income after adjustments
Average capital employed
Return on Average Capital Employed

Return on Stockholders' Equity (ROSE) Net Stockholders' Equity. Average stockholders' beginning and end of the year. ROSE is a ratio

Millions of dollars
Net income attributable to Chevron
Chevron Corporation Stockholders' Equity at December 3
Average Chevron Corporation Stockholders' Equity
Return on Average Stockholders' Equity

Financial and Derivative Instrument Mar

The market risk associated with the company's estimates of financial exposure to market risk actual impact of future market changes could affect materially due to factors discussed elsewhere in this report, including those set forth under the heading "Risk Factors" in Part I, Item 1A of the company's Annual Report on Form 10-K.



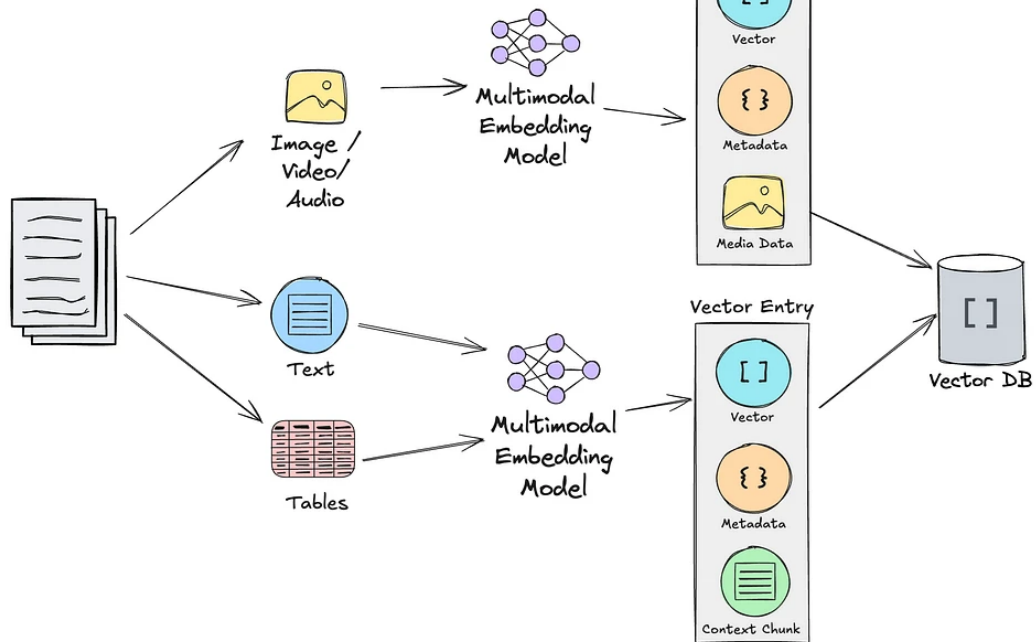
Retrieved Document



vectorize.io

Multimodal RAG Pattern 2

Embed media directly and store raw media in the vector DB



Retrieved Document

The Challenge

MMDocIR-Challenge

Task 1: MMDocIR – Multi-Modal Retrieval for Long Documents

This task is designed to evaluate the ability of retrieval systems to identify visually-rich information within documents. MMDocIR evaluation set includes 313 long documents averaging 65.1 pages, categorized into diverse domains: research reports, administration, industry, tutorials, workshops, academic papers, brochures, financial reports, guidebooks, government documents, laws, and news articles. Different domains feature distinct distributions of multi-modal information.

Start

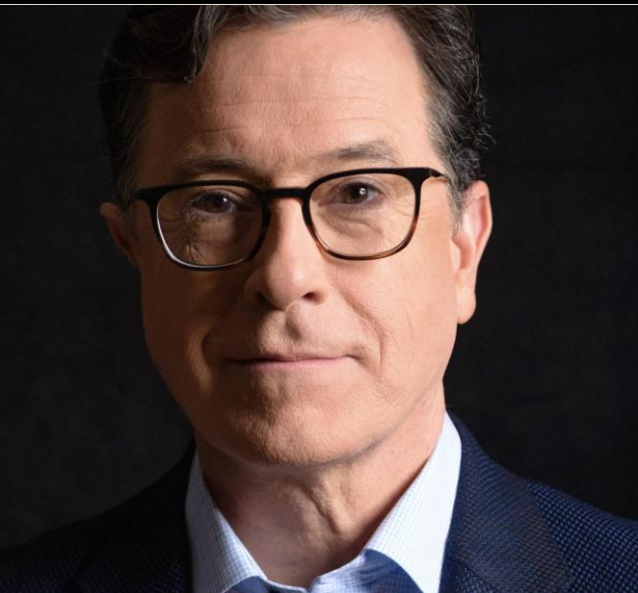
2 months ago

Close

3 days to go



Our approach



ColBERT

Contextualized late interaction BERT



Colpali / ColQwen

Using Vision LLMs + late interaction to improve document retrieval (RAG, search engines, etc.), solely using the image representation of document pages

USING COMPLEX RETRIEVAL SYSTEMS THAT RELY ON OCR, DOCUMENT LAYOUT RECOGNITION, CHUNKING STRATEGIES, FIGURE CAPTIONING AND POWERFUL TEXT EMBEDDING MODELS

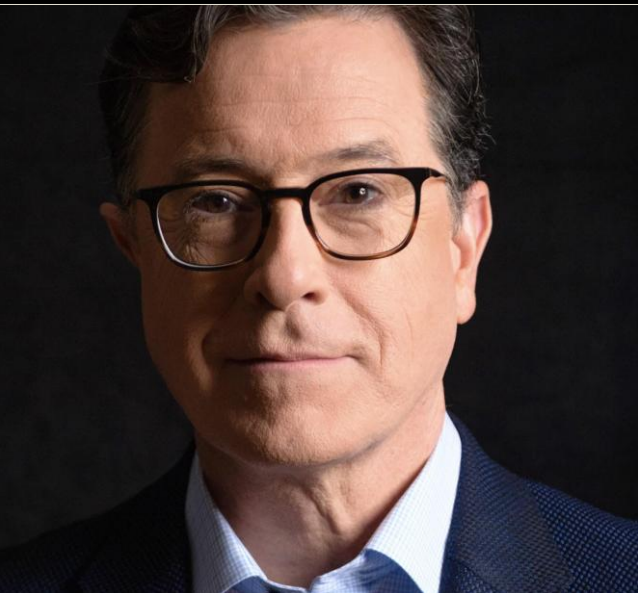
JUST EMBED THE IMAGE



Less than vanilla RAG

Only algorithms, no language model or embedding.

Our approach



ColBERT

DID NOT WORK! Memory issues.



USING COMPLEX RETRIEVAL SYSTEMS
THAT RELY ON OCR, DOCUMENT LAYOUT
RECOGNITION, CHUNKING STRATEGIES,
FIGURE CAPTIONING AND POWERFUL
TEXT EMBEDDING MODELS

JUST EMBED THE IMAGE

Colpali / ColQwen

DID NOT WORK! Memory issues.

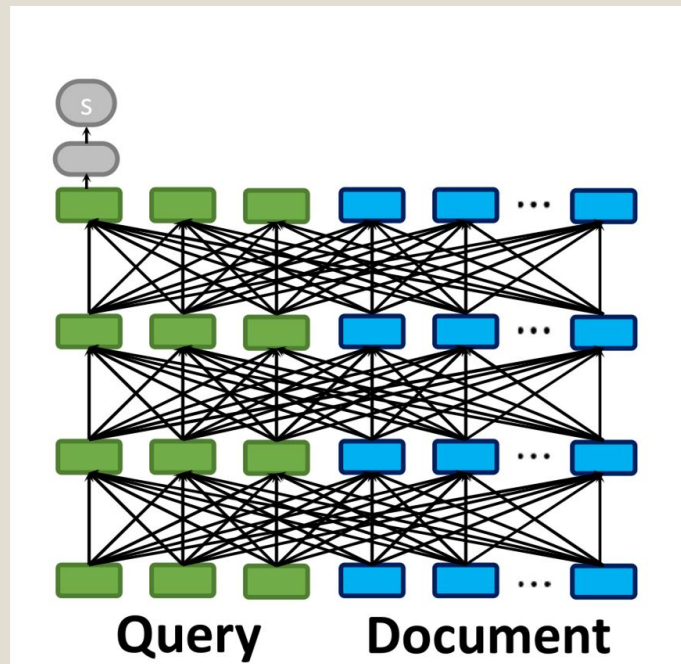


Less than vanilla RAG

WORKED! Provided output, however poor
the quality

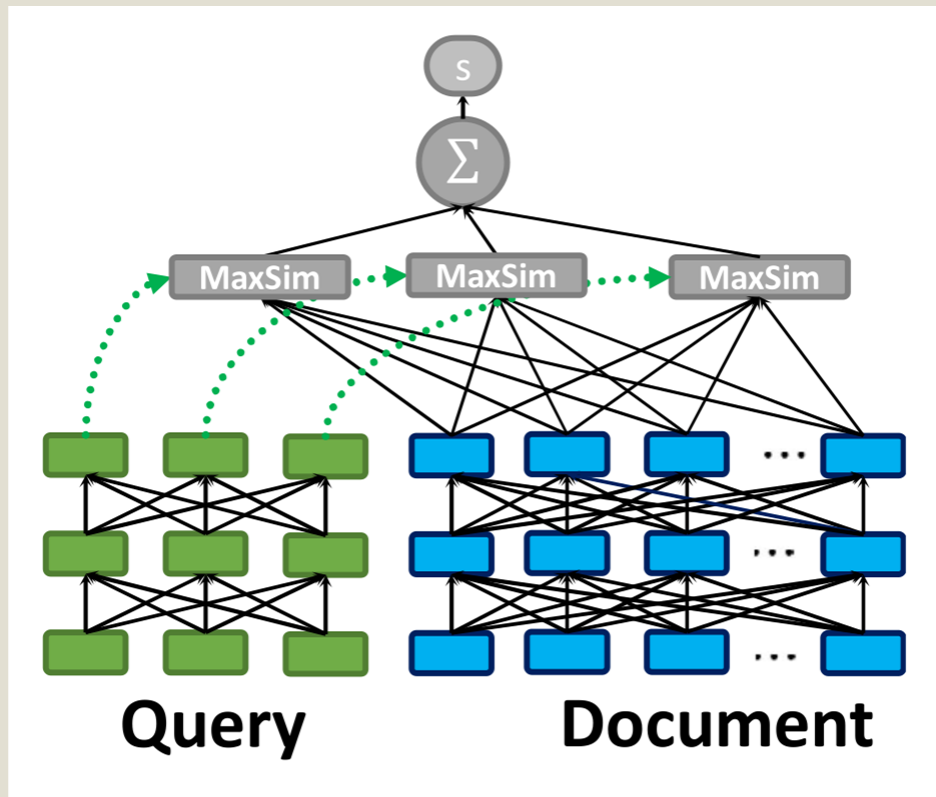
A note on BERT

- Bidirectional context.
- Masked Language Model
- Next Sentence Prediction



"All-to-all Interaction"

Co1BERT

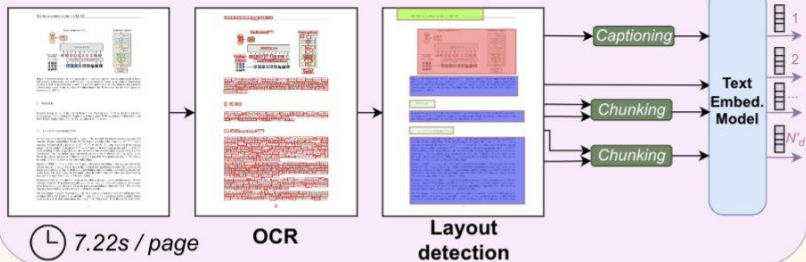


"Late Interaction"

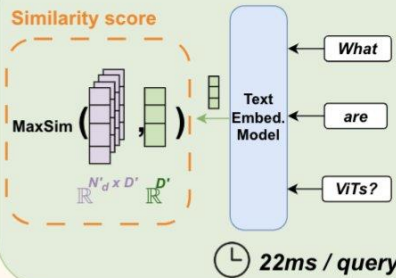
ColPali

Standard Retrieval 📊 0.66 NDCG@5

offline

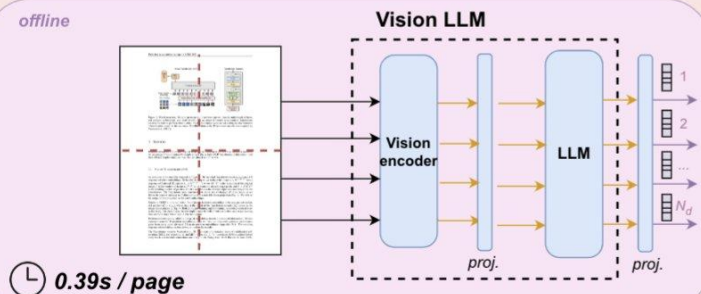


online



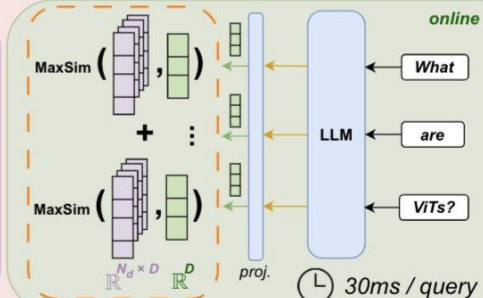
ColPali (ours) 📊 0.81 NDCG@5

offline

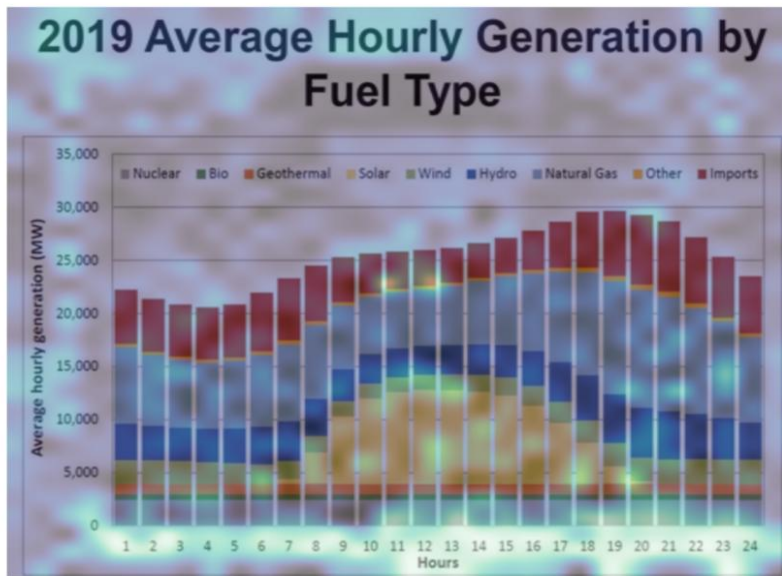


Similarity score

online

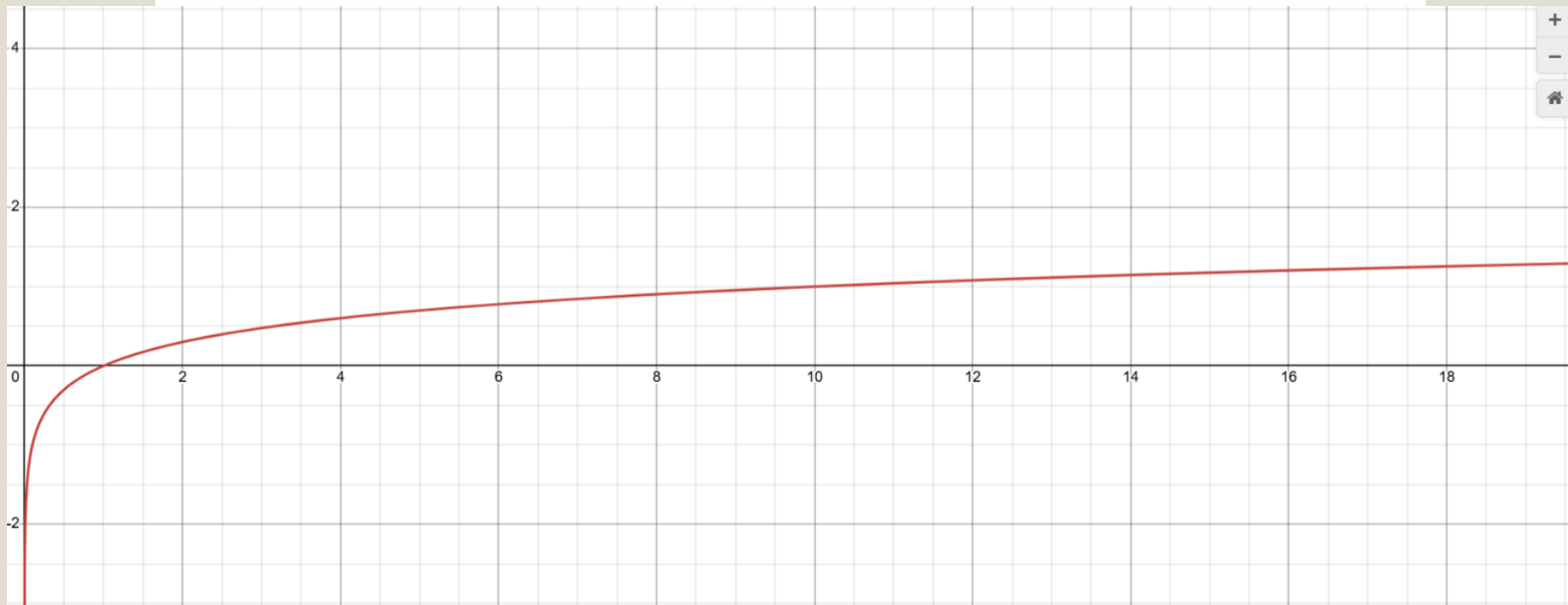


ColPali



Query: "Which hour of the day had the highest overall electricity generation in 2019?"

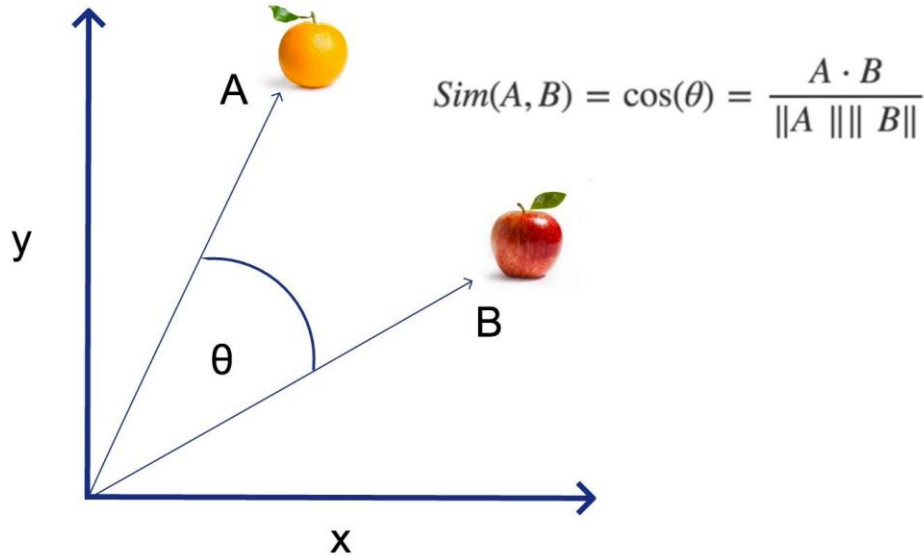
vanilla deep dive: TF-IDF



J' # at p with t /

vanilla deep dive: cosine similarity

Cosine Similarity



Results

TF-IDF



```
scores = {}
for question_id, question, doc_name in zip(df_gt['question_id'], df_gt['question'], df_gt['doc_name']):
    document = df_passages[df_passages['doc_id'] == doc_name]['vlm_text'].tolist()
    terms = question.split()
    passage_scores = []

    for i, page in enumerate(document):
        total_score = sum(term_frequency(t, page) * inverse_document_frequency(t, document) for t in terms)
        passage_scores.append((i, total_score))

    scores[question_id] = passage_scores

df_scores = pd.DataFrame([
    {"question_id": qid, "page_index": idx, "score": score}
    for qid, passages in scores.items() for idx, score in passages
])
```

⊗ 160m 48.9s

KeyboardInterrupt Traceback (most recent call last)

cell In[96], line 10

```
7 passage_scores = []
9 for i, page in enumerate(document):
--> 10     total_score = sum(term_frequency(t, page) * inverse_document_frequency(t, document) for t in terms)
11     passage_scores.append((i, total_score))
13 #top_5_passages = heapq.nlargest(5, passage_scores, key=lambda x: x[1])
14 #scores[question_id] = top_5_passages
```

THANK YOU

Sources

<https://docs.datastax.com/en/ragstack/colbert/index.html>

<https://huggingface.co/blog/manu/colpali>

<https://www.kaggle.com/competitions/multimodal-document-retrieval-challenge>